fatigue properties due to continual exposure to hydrogen , as necessary.

Hydrogen embrittlement (see 4.3)

A-7 Double tube structures ensuring no leakage, or fixed hydrogen detectors being capable of detecting a hydrogen leak, should be provided for confined places where leakage of hydrogen may occur, such as cargo valves, flanges and seals.

Susceptibility to leakage (see 4.4.2)

A-8 Helium or a mixture of 5% hydrogen and 95% nitrogen should be used as the tightness test medium for cargo tank and cargo piping.

Susceptibility to leakage (see 4.4.3)

A-9 The amount of carbon dioxide carried for a carbon dioxide fire-extinguishing system should be sufficient to provide a quantity of free gas equal to 75% or more of the gross volume of the cargo compressor and pump rooms in all cases.

Fire by Hydrogen (see 4.7.3)

Wide range of

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the cargo compressor and pump rooms in all cases.

Fire by Hydrogen (see 4.7.3)

Wide range of

flammability limits (see 4.10)

A-10 When deterioration of insulation capability by single damage is possible, appropriate safety measures should be adopted taking into account the deterioration.

High pressure (see 4.8)

A-11 Appropriate measures should be provided to prevent vents becoming blocked by accumulations of ice formed from moisture in the air.

Low temperature (see 4.2.2)

A-12 Due consideration should be given to means for handling boil-off gas.

High pressure (see 4.8)

A-13 Due consideration should be given to static electricity associated with rotating or reciprocating machinery including the installation of conductive machinery belts and precautionary measures incorporated in operating and maintenance procedures , in

addition to the bonding of tanks, piping and equipment required
by paragraph 5.7.4 of the Code.

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addition to the bonding of tanks, piping and equipment required
by paragraph 5.7.4 of the Code.
Anti-static clothing and footwear, and a portable hydrogen
detector should be provided for each crew member working in
the cargo area.
Static electricity
(see 4.9.2)
A-14 The cargo operation manuals required in paragraph 18.2 of the
Code should include limitations of various operations in relation
to environmental conditions.
Wide range of
flammability limits
(see 4.10)

No. Special requirement Related hazard
A-15 An appropriate procedure should be established for warm -up,
inert gas purge, gas -free, hydrogen purge and pre -cooling.
The procedure should include:
.1 selection of inert gas in relation to temperature limit;
.2 measurement of gas concentration;

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The procedure should include:
.1 selection of inert gas in relation to temperature limit;
.2 measurement of gas concentration;
.3 measurement of temperature;
.4 rates of supply of gases;
.5 conditions for commencement, suspension, resuming
and termination of each operation;
.6 treatment of return gases; and
.7 discharge of gases.
Prevention of
dangerous
purging operation
(see 4.11)
A-16 Only almost pure para-hydrogen (i.e. more than 95%) should be
loaded in order to avoid excessive heating by ortho - to
para-hydrogen conversion.
General

(see 4.1)
A-17 Fire detectors for detecting hydrogen fire should be selected, taking into account the features of hydrogen fire, to the satisfaction of the Administration.
Features of hydrogen fire and fire hazard
(see 4.7.4)
A-18 At the design stage, dispersion of hydrogen from vent outlets should be analysed in order to minimize risk of ingress of flammable gas into accommodation spaces, service spaces,
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should be analysed in order to minimize risk of ingress of flammable gas into accommodation spaces, service spaces,
machinery spaces and control stations. Extension of hazardous areas should be considered based on the results of the analysis.
Low density and high diffusivity
(see 4.5)
A-19 Due consideration should be given to appropriate safety measures to prevent formation of explosive mixture in the case of a leakage and permeation of hydrogen, including:
.1 installation of hydrogen detectors in order to detect a possible ground -level travel of low temperature hydrogen gas, and at high points in spaces where warm hydrogen gas can be trapped; and
.2 application of "best practice" for land -based liquid hydrogen storage taking into account appropriate guidance such as "Cryogenics Safety Manual Fourth Edition (1998)"8).
General
(see 4.1)
A-20 In the case that fusible elements are used as a means of fire
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SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
Edition (1998)"8).
General
(see 4.1)
A-20 In the case that fusible elements are used as a means of fire detection required by paragraph 18.10.3.2 of the Code, flame detectors suitable for hydrogen flames should be provided in addition at the same locations. Appropriate means should be adopted to prevent the activation of ESD system owing to false alarm of flame detectors, e.g. avoiding activation of ESD system by single sensor (voting method).
Fire hazard
(see 4.7.4)

No. Special requirement Related hazard
A-21 Consideration should be given to enhance the ventilation capacity of the enclosed spaces subject to liquefied hydrogen leakage and permeation, taking into account the latent heat of vaporization, specific heat and the volume of hydrogen gas in
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leakage and permeation, taking into account the latent heat of vaporization, specific heat and the volume of hydrogen gas in relation to temperature and heat capacity of adjacent spaces.
Low density and high diffusivity (see 4.5)
A-22 Liquid and gas hydrogen pipes should not pass through enclosed spaces in addition to other than those referred to in paragraph 5.2.2.1.2 of the Code, unless:
.1.1 the spaces are equipped with gas detection systems which activate the alarm at not more than 20% LFL and shut down the isolation valves, as appropriate, at not more than 40% LFL (see sections 16.4.2 and 16.4.8 of the Code); and
.1.2 the spaces are adequately ventilated; or
.2 the spaces are maintained in an inert condition.
Susceptibility to leakage (see 4.4)
A-23 A risk assessment should be conducted to ensure that risks